

Perform Network Analysis Using the NetworkMiner Tool

1. AIM

To perform network analysis using the NetworkMiner tool and identify useful network forensic information from a packet-capture (PCAP/PCAPNG) file.

2. OBJECTIVES

- Understand NetworkMiner and its purpose.
- Install and open NetworkMiner on Windows.
- Understand its main tabs and interface.
- Load a PCAP/PCAPNG file.
- Identify hosts, IP addresses and available host information.
- Examine sessions and communication information.
- Examine DNS/domain information where available.
- Examine files, images and other supported artifacts.
- Record observations in a lab record.

3. INTRODUCTION TO NETWORKMINER

NetworkMiner is a Network Forensic Analysis Tool (NFAT). It helps an analyst extract and organize useful information from captured network traffic. Instead of manually reading every packet, NetworkMiner presents available information in views such as Hosts, Files, Images, Sessions and DNS.

NetworkMiner commonly works with packet-capture files such as PCAP and PCAPNG. When a capture is loaded, it analyzes the recorded packets and attempts to extract information that can help an investigator understand communicating systems, protocols and supported artifacts.

3.1 What is Network Forensic Analysis?

Network forensic analysis is the examination of network traffic to understand communication and identify useful evidence such as hosts, protocols, domains, sessions and transferred artifacts.

3.2 Why NetworkMiner is Useful

- Easy-to-understand forensic views.
- Host and IP identification.
- Session and communication review.
- Supported file and image extraction.
- DNS/domain information review.
- Faster review of a PCAP than manually examining every packet.

4. BASIC NETWORK CONCEPTS

IP Address

Identifies a device or network interface at the IP layer. Example: 192.168.1.10.

MAC Address

A hardware/interface address used at the data-link layer.

Port

Identifies a service endpoint used by TCP or UDP. Example: 80 and 443.

Protocol

Rules used for communication, such as TCP, UDP, DNS, HTTP and HTTPS.

PCAP / PCAPNG

Packet-capture formats containing recorded network packets and related metadata.

5. SOFTWARE AND HARDWARE REQUIREMENTS

- Windows 10/11 computer.
- NetworkMiner.
- Authorized sample PCAP/PCAPNG file.
- Optional: Wireshark for creating/reviewing a lab capture.
- Sufficient disk space.

6. INSTALLING NETWORKMINER

Use the official NetworkMiner source approved by your institution. NetworkMiner is commonly distributed as a downloadable archive.

1. Open Chrome or Edge.
2. Search for the official NetworkMiner download page.
3. Download the Windows package.
4. Locate the downloaded ZIP/archive.
5. Right-click it and choose Extract All.
6. Extract it to a folder such as C:\NetworkMiner.
7. Open the extracted folder.
8. Locate the NetworkMiner executable.
9. Run NetworkMiner using the permissions specified by your lab.
10. Wait for the main window to open.

Important: Do not download forensic software from unknown websites.

7. OPENING NETWORKMINER

11. Open the NetworkMiner folder.
12. Double-click the NetworkMiner executable.
13. Follow the Windows security prompt according to your lab policy.
14. Wait for the main interface to appear.
15. For student work, use only an instructor-provided or intentionally generated PCAP.

8. UNDERSTANDING THE NETWORKMINER INTERFACE

The exact tabs can vary between versions. Students should record only information actually displayed by their installed version.

View	Purpose
Hosts	Shows identified hosts and available IP, hostname, MAC/vendor, OS and related information.
Files	Shows files identified or reconstructed from supported traffic.
Images	Shows images reconstructed from suitable traffic.
Sessions / Messages	Shows available communication/session information.
DNS	Shows DNS-related information available in the capture.
Credentials / Related Artifacts	Some versions/traffic may expose authentication-related information. Use only authorized lab traffic; encryption may prevent plaintext credentials from being visible.

9. PRACTICAL PREPARATION – SAMPLE PCAP

For a college laboratory, use a sample PCAP supplied by the instructor or intentionally generated in the lab. Do not inspect other people's private network traffic without authorization.

16. Create C:\NetworkMiner_Lab.
17. Copy the authorized PCAP/PCAPNG into this folder.
18. Use a simple filename such as Lab_Network_Traffic.pcap.
19. Keep the original PCAP unchanged.
20. Start NetworkMiner.

10. LOADING A PCAP

21. Open NetworkMiner.
22. Use the File/Open or load option available in your version.
23. Select the sample PCAP/PCAPNG.
24. Click Open.
25. Wait while NetworkMiner parses the capture.
26. Start the examination with the Hosts tab.

11. ANALYSIS – HOSTS TAB

The Hosts tab is one of the most important views. It helps identify systems appearing in the captured communication.

27. Click Hosts.
28. Observe IP addresses.
29. Check available hostnames.
30. Check MAC/vendor information if shown.
31. Check operating-system/device information if shown.
32. Identify likely client/server roles using the available evidence, not assumptions.
33. Record important host information.

Example

If the tool displays 192.168.1.10 and 192.168.1.20, record them exactly as shown. Do not assume a role only from the IP address.

12. ANALYSIS – FILES TAB

The Files tab may show files identified or reconstructed from supported traffic. Availability depends on the protocols and contents in the PCAP.

34. Click Files.
35. Observe whether files are listed.
36. Record filename, type, source and destination when available.
37. Do not execute recovered files.
38. Keep recovered evidence in the designated lab folder.

13. ANALYSIS – IMAGES TAB

The Images tab may show images reconstructed from suitable traffic. Encrypted HTTPS traffic normally prevents simple recovery of application content from a passive capture.

39. Click Images.
40. Check whether images were recovered.
41. Record image name or available metadata.
42. If none are shown, record that no images were recovered.

14. ANALYSIS – DNS INFORMATION

DNS translates domain names to IP addresses. DNS information can help show which domains a host attempted to resolve.

43. Open the DNS-related view if available.
44. Observe domain names/queries.
45. Record the associated host when shown.
46. Record resolved addresses if shown.
47. Compare DNS observations with Hosts information.

15. ANALYSIS – SESSIONS / COMMUNICATION

Sessions represent communication between systems. The exact presentation depends on the NetworkMiner version and protocols in the PCAP.

48. Open Sessions, Messages or the equivalent view.
49. Observe source and destination hosts.
50. Observe protocol information.
51. Observe ports when shown.
52. Record significant communication pairs.
53. Do not access systems identified in a PCAP; analyze only the supplied evidence.

16. UNDERSTANDING ENCRYPTED TRAFFIC

NetworkMiner cannot automatically decrypt modern encrypted traffic. HTTPS/TLS normally protects application content, although metadata such as IP addresses, ports and timing may still be visible depending on the protocol and capture.

17. COMPLETE PRACTICAL FLOW

PCAP → Open in NetworkMiner → Hosts → Sessions → DNS → Files → Images → Other available artifacts → Observations → Result

18. OBSERVATION TABLE

Parameter	Observation
NetworkMiner Version	
PCAP Filename	
PCAP/PCAPNG Type	
Number of Hosts	
Client IP	
Server/Destination IP	
Hostname(s)	
MAC/Vendor	
Operating System	
Protocols	
Ports	
DNS/Domains	
Files Recovered	
Images Recovered	
Sessions/Communication	
Other Findings	

19. SCREENSHOTS FOR RECORD

54. NetworkMiner main window.
55. Hosts tab.
56. Files tab, if files are available.
57. Images tab, if images are available.

58. DNS/session-related view, if available.

59. Final analysis/result screen.

If a tab has no data, a screenshot of the empty result may be taken and the observation can state: 'No artifact available in the supplied PCAP.'

20. RESULT

Thus, the given network packet-capture file was successfully analyzed using NetworkMiner. Available hosts, communication information, DNS details, files, images and other network forensic artifacts were identified and recorded.

21. PRECAUTIONS

- Use NetworkMiner only for authorized forensic/educational analysis.
- Use instructor-provided or intentionally generated PCAPs.
- Do not capture or inspect private traffic without authorization.
- Do not execute recovered files.
- Keep the original PCAP unchanged.
- Do not assume every tab will contain data.
- Remember that encryption may hide application contents.
- Record observations exactly as shown.

22. COMMON PROBLEMS AND SOLUTIONS

Problem	Solution
NetworkMiner does not start	Check that the package is complete and use the supported Windows environment/permissions.
PCAP does not open	Check path and file validity; try a known-good instructor PCAP.
No hosts shown	The capture may be empty, invalid or unsupported. Try the known-good sample.
No files shown	The PCAP may not contain supported file-transfer/application traffic.
No images shown	Traffic may be encrypted or contain no recoverable images.
Some fields are blank	The capture may not contain enough information to populate those fields.

23. VIVA QUESTIONS AND ANSWERS

Q. What is NetworkMiner?

A. A Network Forensic Analysis Tool used to extract and organize information from network traffic.

Q. What is network forensic analysis?

A. Examination of network traffic to understand communication and identify useful evidence.

Q. What is PCAP?

A. A packet-capture file containing recorded network packets and related information.

Q. What is the Hosts tab?

A. A view used to identify hosts and available host-related information.

Q. What can the Files tab show?

A. Files identified or reconstructed from supported network traffic.

Q. What can the Images tab show?

A. Images recovered from suitable supported traffic.

Q. What is DNS?

A. A system that translates domain names into IP addresses.

Q. Can NetworkMiner decrypt HTTPS?

A. Normally no. HTTPS/TLS protects application content in a passive capture.

Q. Why use a sample PCAP?

A. It provides controlled, authorized evidence for a classroom exercise.

Q. Can every PCAP contain files/images?

A. No. Results depend on traffic, protocols, contents and encryption.

Q. Why preserve the original PCAP?

A. To maintain evidence integrity and allow analysis to be repeated.

Q. What does NFAT mean?

A. Network Forensic Analysis Tool.

24. RECORD WRITING – SHORT FORMAT

Aim

To perform network analysis using the NetworkMiner tool and identify useful network forensic information from a packet-capture file.

Procedure

60. Opened NetworkMiner.
61. Obtained an authorized sample PCAP/PCAPNG.
62. Loaded the capture into NetworkMiner.
63. Examined Hosts.
64. Examined sessions/communication.
65. Examined DNS information where available.
66. Examined Files and Images.
67. Recorded observations and screenshots.

Result

The network packet capture was successfully analyzed using NetworkMiner and the available network forensic artifacts were identified and documented.

25. CONCLUSION

NetworkMiner provides a convenient way to organize information extracted from packet captures. It helps students understand how raw network traffic can be converted into forensic observations such as hosts, sessions, DNS information, files and images. The exact artifacts depend on the contents and encryption of the capture.